

SQL Based Data Analysis: Query Outputs and Results

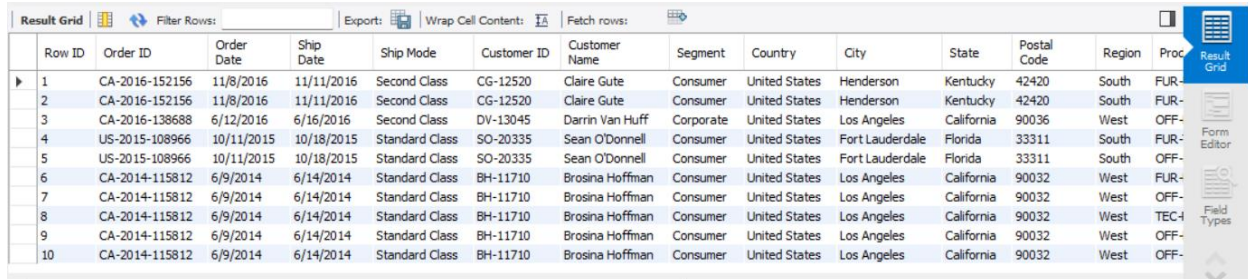
```
CREATE DATABASE SuperstoreDB;
```

```
USE SuperstoreDB;
```

1. Explore Table

View first 10 rows

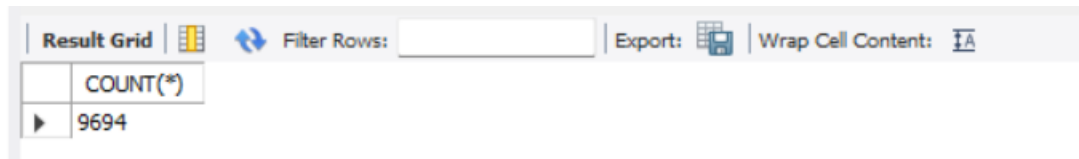
```
SELECT * FROM superstore LIMIT 10;
```



Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product Name
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TECH
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-

Total Number of records

```
SELECT COUNT(*) FROM superstore;
```



COUNT(*)
9694

--View table schema

```
DESCRIBE superstore;
```

Result Grid						
Filter Rows:		Exports:		Wrap Cell Content:		
	Field	Type	Null	Key	Default	Extra
▶	Row ID	int	YES		NULL	
	Order ID	text	YES		NULL	
	Order Date	text	YES		NULL	
	Ship Date	text	YES		NULL	
	Ship Mode	text	YES		NULL	
	Customer ID	text	YES		NULL	
	Customer Name	text	YES		NULL	
	Segment	text	YES		NULL	
	Country	text	YES		NULL	
	City	text	YES		NULL	
	State	text	YES		NULL	
	Postal Code	int	YES		NULL	
	Region	text	YES		NULL	
	Product ID	text	YES		NULL	
	Category	text	YES		NULL	
	Sub-Category	text	YES		NULL	
	Product Name	text	YES		NULL	
	Sales	double	YES		NULL	
	Quantity	int	YES		NULL	
	Discount	double	YES		NULL	
	Profit	double	YES		NULL	

2. WHERE FILTERS

--Orders from West region

SELECT *

FROM superstore

WHERE Region='West';

Result Grid														
Filter Rows:		Exports:		Wrap Cell Content:		Fetch rows:								
	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Pr
▶	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OF
	6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FU
	7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OF
	8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TE
	9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OF
	10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OF
	11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FU
	12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TE
	14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OF
	18	CA-2014-167164	5/13/2014	5/15/2014	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West	OF
	19	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OF
	20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	TE
	21	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OF
	25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FU
	26	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	OF
	27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TE
	43	CA-2016-101343	7/17/2016	7/22/2016	Standard Class	RA-19885	Ruben Ausman	Corporate	United States	Los Angeles	California	90049	West	OF
	63	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	TE
	64	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	OF
	65	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	OF

--Furniture category orders

SELECT *

FROM superstore

WHERE Category='Technology';

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West
20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West
36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central
41	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	United States	Houston	Texas	77041	Central
42	CA-2017-120999	9/10/2017	9/15/2017	Standard Class	LC-16930	Linda Cazamas	Corporate	United States	Naperville	Illinois	60540	Central
45	CA-2016-118255	3/11/2016	3/13/2016	First Class	ON-18715	Odella Nelson	Corporate	United States	Eagan	Minnesota	55122	Central
48	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United States	Dover	Delaware	19901	East
49	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United States	Dover	Delaware	19901	East
55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate	United States	New York City	New York	10024	East
60	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United States	Troy	New York	12180	East
63	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West
69	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brendan Sweed	Corporate	United States	Gilbert	Arizona	85234	West
87	CA-2017-155558	10/26/2017	11/2/2017	Standard Class	PG-18895	Paul Gonzalez	Consumer	United States	Rochester	Minnesota	55901	Central
91	CA-2016-109806	9/17/2016	9/22/2016	Standard Class	JS-15685	Jim Sink	Corporate	United States	Los Angeles	California	90036	West
101	CA-2016-158568	8/29/2016	9/2/2016	Standard Class	RB-19465	Rick Bensley	Home Office	United States	Chicago	Illinois	60610	Central
104	US-2015-156867	11/13/2015	11/17/2015	Standard Class	LC-16870	Lena Cacioppo	Consumer	United States	Aurora	Colorado	80013	West
107	CA-2017-119004	11/23/2017	11/28/2017	Standard Class	JM-15250	Janet Martin	Consumer	United States	Charlotte	North Ca...	28205	South
108	CA-2017-119004	11/23/2017	11/28/2017	Standard Class	JM-15250	Janet Martin	Consumer	United States	Charlotte	North Ca...	28205	South

--Sales Greater Than 1000

SELECT *

FROM superstore

WHERE Sales > 1000;

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East
36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central
55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate	United States	New York City	New York	10024	East
68	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brendan Sweed	Corporate	United States	Gilbert	Arizona	85234	West
150	CA-2016-114489	12/5/2016	12/9/2016	Standard Class	JE-16165	Justin Ellison	Corporate	United States	Franklin	Wisconsin	53132	Central
166	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	Texas	78207	Central
168	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	Texas	78207	Central
216	CA-2015-146262	1/2/2015	1/9/2015	Standard Class	VW-21775	Victoria Wilson	Corporate	United States	Medina	Ohio	44256	East
245	CA-2014-131926	6/1/2014	6/6/2014	Second Class	DW-13480	Dianna Wilson	Home Off...	United States	Lakeville	Minnesota	55044	Central
248	CA-2014-131926	6/1/2014	6/6/2014	Second Class	DW-13480	Dianna Wilson	Home Off...	United States	Lakeville	Minnesota	55044	Central
252	CA-2016-145625	9/11/2016	9/17/2016	Standard Class	KC-16540	Kelly Collier	Consumer	United States	San Diego	California	92037	West
263	US-2014-106992	9/19/2014	9/21/2014	Second Class	SB-20290	Sean Braxton	Corporate	United States	Houston	Texas	77036	Central
264	US-2014-106992	9/19/2014	9/21/2014	Second Class	SB-20290	Sean Braxton	Corporate	United States	Houston	Texas	77036	Central
282	US-2015-161991	9/26/2015	9/28/2015	Second Class	SC-20725	Steven Cartwright	Consumer	United States	Houston	Texas	77070	Central
283	CA-2015-130890	11/2/2015	11/6/2015	Standard Class	JO-15280	Jas O'Carroll	Consumer	United States	Los Angeles	California	90004	West
319	CA-2014-164973	11/4/2014	11/9/2014	Standard Class	NM-18445	Nathan Mautz	Home Off...	United States	New York City	New York	10024	East
354	CA-2016-129714	9/1/2016	9/3/2016	First Class	AB-10060	Adam Bellavance	Home Off...	United States	New York City	New York	10009	East
370	CA-2016-155516	10/21/2016	10/21/2016	Same Day	MK-17905	Michael Kennedy	Corporate	United States	Manchester	Connecticut	6040	East

--Orders Placed

SELECT *

FROM superstore

WHERE `Order Date` LIKE '%2016%';

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Prc
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF
14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF
22	CA-2016-137330	12/9/2016	12/13/2016	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF
23	CA-2016-137330	12/9/2016	12/13/2016	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF
26	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	OFF
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC
36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	TEC
37	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	FUR
43	CA-2016-101343	7/17/2016	7/22/2016	Standard Class	RA-19885	Ruben Ausman	Corporate	United States	Los Angeles	California	90049	West	OFF

3. AGGREGATION FUNCTIONS

--Total Sales

```
SELECT SUM(Sales) AS Total_Sales
FROM superstore;
```

Total_Sales
2272449.8562999545

--Total Quantity Sold

```
SELECT SUM(Quantity) AS Total_Quantity
FROM superstore;
```

Total_Quantity
36749

--Average Sales

```
SELECT AVG(Sales) AS Average_Sales
FROM superstore;
```

Average_Sales
234.41818199917006

--Maximum Sale

```
SELECT MAX(Sales) AS Highest_Sale
FROM superstore;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Highest_Sale			
22638.48			

--Minimum Sales

```
SELECT MIN(Sales) AS Lowest_Sale
FROM superstore;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Lowest_Sale			
0.444			

4. Sort and Limit results

--Top 10 Products by Sales

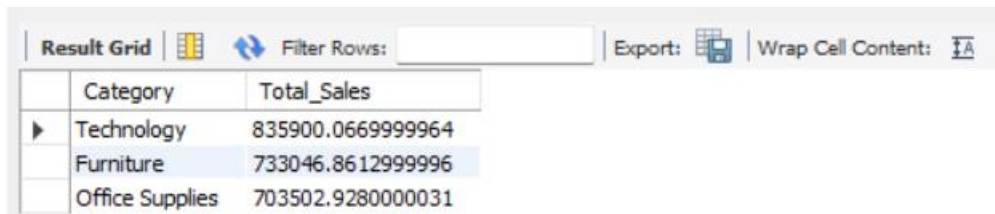
```
SELECT `Product Name`,
       SUM(Sales) AS Total_Sales
FROM superstore
GROUP BY `Product Name`
ORDER BY Total_Sales DESC
LIMIT 10;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
Product Name	Total_Sales			
Canon imageCLASS 2200 Advanced Copier	61599.824			
Fellowes PB500 Electric Punch Plastic Comb Bind...	27453.384			
Cisco TelePresence System EX90 Videoconferen...	22638.48			
HON 5400 Series Task Chairs for Big and Tall	21870.576			
GBC DocuBind TL300 Electric Binding System	19823.479000000003			
GBC Ibimaster 500 Manual ProClick Binding System	19024.5			
Hewlett Packard LaserJet 3310 Copier	18839.686			
HP Designjet T520 Inkjet Large Format Printer -...	18374.895			
GBC DocuBind P400 Electric Binding System	17965.068			
High Speed Automatic Electric Letter Opener	17030.311999999998			

--Top categories by Sales

```
SELECT Category,
       SUM(Sales) AS Total_Sales
FROM superstore
GROUP BY Category
```

ORDER BY Total_Sales DESC;

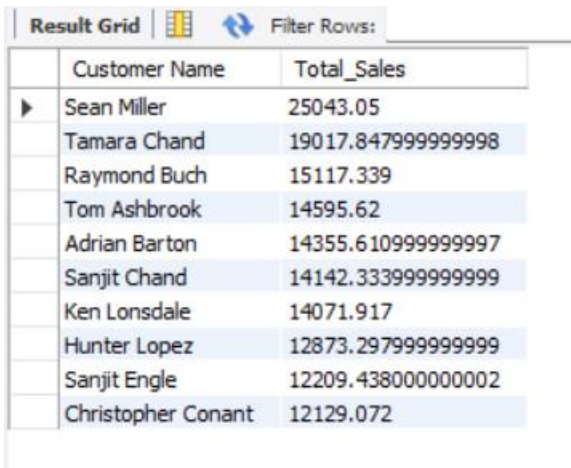


The screenshot shows a database interface with a 'Result Grid' tab. It includes a 'Filter Rows' search bar, an 'Export' button with a grid icon, and a 'Wrap Cell Content' toggle. The table below lists the top categories by total sales.

	Category	Total_Sales
▶	Technology	835900.0669999964
	Furniture	733046.8612999996
	Office Supplies	703502.9280000031

--Top 10 Customers by Sales

```
SELECT `Customer Name`,  
       SUM(Sales) AS Total_Sales  
FROM superstore  
GROUP BY `Customer Name`  
ORDER BY Total_Sales DESC  
LIMIT 10;
```



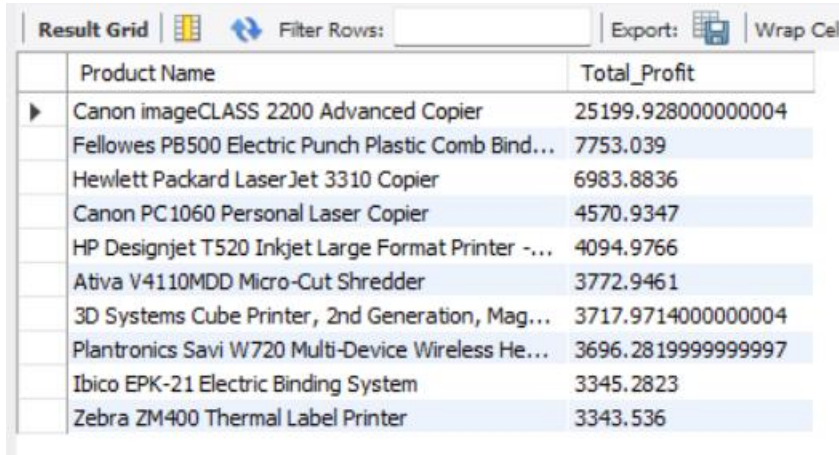
The screenshot shows a database interface with a 'Result Grid' tab. It includes a 'Filter Rows' search bar. The table below lists the top 10 customers by total sales.

	Customer Name	Total_Sales
▶	Sean Miller	25043.05
	Tamara Chand	19017.847999999998
	Raymond Buch	15117.339
	Tom Ashbrook	14595.62
	Adrian Barton	14355.610999999997
	Sanjit Chand	14142.333999999999
	Ken Lonsdale	14071.917
	Hunter Lopez	12873.297999999999
	Sanjit Engle	12209.438000000002
	Christopher Conant	12129.072

--Top 10 Most Profitable Products

```
SELECT `Product Name`,  
       SUM(Profit) AS Total_Profit  
FROM superstore  
GROUP BY `Product Name`  
ORDER BY Total_Profit DESC
```

LIMIT 10;



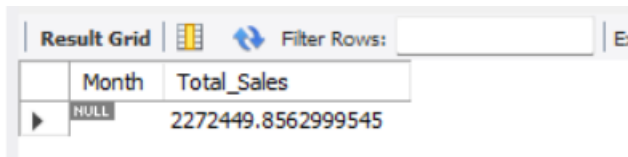
The screenshot shows a 'Result Grid' window with a toolbar containing 'Filter Rows', 'Export', and 'Wrap Cell' options. The grid displays a list of products and their total profits, sorted in descending order. The first product, 'Canon imageCLASS 2200 Advanced Copier', has a total profit of 25199.928000000004.

Product Name	Total_Profit
Canon imageCLASS 2200 Advanced Copier	25199.928000000004
Fellowes PB500 Electric Punch Plastic Comb Bind...	7753.039
Hewlett Packard LaserJet 3310 Copier	6983.8836
Canon PC1060 Personal Laser Copier	4570.9347
HP Designjet T520 Inkjet Large Format Printer -...	4094.9766
Ativa V4110MDD Micro-Cut Shredder	3772.9461
3D Systems Cube Printer, 2nd Generation, Mag...	3717.9714000000004
Plantronics Savi W720 Multi-Device Wireless He...	3696.2819999999997
Ibico EPK-21 Electric Binding System	3345.2823
Zebra ZM400 Thermal Label Printer	3343.536

5. SOLVE USE CASES

--Monthly Sales Trend

```
SELECT DATE_FORMAT(`Order Date`, '%Y-%m') AS Month,  
       SUM(Sales) AS Total_Sales  
FROM superstore  
GROUP BY DATE_FORMAT(`Order Date`, '%Y-%m')  
ORDER BY Month;
```

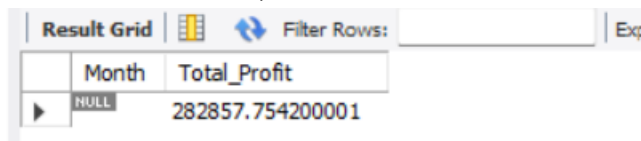


The screenshot shows a 'Result Grid' window with a toolbar containing 'Filter Rows' and 'Export' options. The grid displays a single row for the month of January (represented as '01-01') with a total sales value of 2272449.8562999545.

Month	Total_Sales
01-01	2272449.8562999545

--Monthly Profit Trend

```
SELECT DATE_FORMAT(`Order Date`, '%Y-%m') AS Month,  
       SUM(Profit) AS Total_Profit  
FROM superstore  
GROUP BY DATE_FORMAT(`Order Date`, '%Y-%m')  
ORDER BY Month;
```



The screenshot shows a 'Result Grid' window with a toolbar containing 'Filter Rows' and 'Export' options. The grid displays a single row for the month of January (represented as '01-01') with a total profit value of 282857.754200001.

Month	Total_Profit
01-01	282857.754200001

-- Top 10 Customers by Sales

```
SELECT `Customer Name`,  
       SUM(Sales) AS Total_Sales  
FROM superstore
```

```
GROUP BY `Customer Name`
ORDER BY Total_Sales DESC
LIMIT 10;
```

Result Grid   Filter Rows:		
	Customer Name	Total_Sales
▶	Sean Miller	25043.05
	Tamara Chand	19017.847999999998
	Raymond Buch	15117.339
	Tom Ashbrook	14595.62
	Adrian Barton	14355.610999999997
	Sanjit Chand	14142.333999999999
	Ken Lonsdale	14071.917
	Hunter Lopez	12873.297999999999
	Sanjit Engle	12209.438000000002
	Christopher Conant	12129.072

```
-- Top 10 Customers by Profit
SELECT `Customer Name`,
       SUM(Profit) AS Total_Profit
FROM superstore
GROUP BY `Customer Name`
ORDER BY Total_Profit DESC
LIMIT 10;
```

Result Grid   Filter Rows:		
	Customer Name	Total_Profit
▶	Tamara Chand	8964.482600000001
	Raymond Buch	6976.0959
	Sanjit Chand	5757.411899999999
	Hunter Lopez	5622.4292000000005
	Adrian Barton	5438.9075
	Tom Ashbrook	4703.788299999999
	Christopher Martinez	3899.8903999999998
	Keith Dawkins	3038.6254
	Andy Reiter	2884.6207999999997
	Daniel Raglin	2869.0760000000005

```
-- Duplicate Order IDs
SELECT `Order ID`,
       COUNT(*) AS Occurrences
FROM superstore
GROUP BY `Order ID`
HAVING COUNT(*) > 1;
```

Result Grid			Filter Rows:
	Order ID	Occurrences	
▶	CA-2016-152156	2	
	US-2015-108966	2	
	CA-2014-115812	7	
	US-2015-118983	2	
	CA-2014-143336	3	
	CA-2016-137330	2	
	CA-2016-121755	2	
	US-2015-150630	6	
	CA-2016-117590	2	
	CA-2015-117415	4	
	CA-2016-118255	2	
	CA-2016-169194	2	
	CA-2015-115742	4	
	CA-2016-105816	2	
	CA-2016-105816	2	

Result 36 x

```
-- Number of Duplicate Orders
SELECT COUNT(*) AS Duplicate_Orders
FROM (
    SELECT `Order ID`
    FROM superstore
    GROUP BY `Order ID`
    HAVING COUNT(*) > 1
) AS duplicates;
```

Result Grid		Filter Rows:
	Duplicate_Orders	
▶	2390	

```
--Top 5Customers in Each Region
SELECT Region,
       `Customer Name`,
       SUM(Sales) AS Total_Sales
FROM superstore
GROUP BY Region, `Customer Name`
ORDER BY Region, Total_Sales DESC;
```

Result Grid			
Filter Rows:			
	Region	Customer Name	Total_Sales
▶	Central	Tamara Chand	18402.768
	Central	Adrian Barton	12181.593999999997
	Central	Becky Martin	10539.896
	Central	Sanjit Chand	9900.19
	Central	Harry Marie	6533.558800000001
	Central	Helen Wasserman	6100.746000000001
	Central	Andy Reiter	5802.7
	Central	Sean Braxton	5579.9400000000005
	Central	Tonja Turnell	5301.688
	Central	Laura Armstrong	5076.47
	Central	Alex Avila	4780.552
	Central	Cathy Prescott	4697.37
	Central	Shirley Daniels	4559.684
	Central	Ken Lonsdale	4517.2949999999999
	Central	Philisse Overcash	3910.5849999999996
	Central	Zuschuss Carroll	3869.8059999999996
	Central	Dianna Wilson	3745.63
	Central	Bill Eplett	3501.896
	Central	Heather Kirkland	3300.194

6. Validate results

--Total number of rows

```
SELECT COUNT(*) AS Total_Rows
FROM superstore;
```

Result Grid	
Filter Rows:	
	Total_Rows
▶	9694

--Check total columns

```
DESCRIBE superstore;
```

Result Grid						
		Filter Rows:		Export:		
	Field	Type	Null	Key	Default	Extra
►	Row ID	int	YES		NULL	
	Order ID	text	YES		NULL	
	Order Date	text	YES		NULL	
	Ship Date	text	YES		NULL	
	Ship Mode	text	YES		NULL	
	Customer ID	text	YES		NULL	
	Customer Name	text	YES		NULL	
	Segment	text	YES		NULL	
	Country	text	YES		NULL	
	City	text	YES		NULL	
	State	text	YES		NULL	
	Postal Code	int	YES		NULL	
	Region	text	YES		NULL	
	Product ID	text	YES		NULL	
	Category	text	YES		NULL	
	Sub-Category	text	YES		NULL	
	Product Name	text	YES		NULL	
	Sales	double	YES		NULL	
	Quantity	int	YES		NULL	
	Discount	double	YES		NULL	
	Profit	double	YES		NULL	

--Check for null values

Customer Name

```
SELECT COUNT(*) AS Missing_Customers
FROM superstore
WHERE `Customer Name` IS NULL;
```

Result Grid		Filter Rows:	
Missing_Customers			
►	0		

Product Name

```
SELECT COUNT(*) AS Missing_Products
FROM superstore
WHERE `Product Name` IS NULL;
```

Result Grid		Filter Rows:	
Missing_Products			
►	0		

Sales

```
SELECT COUNT(*) AS Missing_Sales
FROM superstore
```

WHERE Sales IS NULL;

Result Grid	Filter Rows:
Missing_Sales	
0	

-- Check Negative Profit Records

SELECT *

FROM superstore

WHERE Profit < 0;

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South
15	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Centr
16	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Centr
24	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	19140	East
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East
29	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East
37	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Centr
39	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	United States	Houston	Texas	77041	Centr
40	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	United States	Houston	Texas	77041	Centr
67	US-2015-164175	4/30/2015	5/5/2015	Standard Class	PS-18970	Paul Stevenson	Home Office	United States	Chicago	Illinois	60610	Centr
73	US-2015-134026	4/26/2015	5/2/2015	Standard Class	JE-15745	Joel Eaton	Consumer	United States	Memphis	Tennessee	38109	South
75	US-2015-134026	4/26/2015	5/2/2015	Standard Class	JE-15745	Joel Eaton	Consumer	United States	Memphis	Tennessee	38109	South
76	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate	United States	Houston	Texas	77041	Centr

-- Verify Distinct Customers

SELECT COUNT(DISTINCT 'Customer Name') AS Total_Customers

FROM superstore;

Result Grid	Filter Rows:
Total_Customers	
793	

-- Verify Distinct Products

SELECT COUNT(DISTINCT 'Product Name') AS Total_Products

FROM superstore;

Result Grid	Filter Rows:
Total_Products	
1798	